1. Executive Summary

```
1. Executive summary
- Train games=940 | Derby games=75
- Nested CV=5 outer folds x 3 inner folds (expanding time-aware)
- Selected model family: Regime-aware simplex stack (Ridge/Huber/HistGB/GBR/HistGB-bag) + q50 blend
- Selected Elo variant: elo_base_static
- Nested outer RMSE/MAE: 42.727 / 34.160
- Selected postprocess module/eta: expand_regime_affine_v2_heterosk / 0.55
- Calibration=regime | Scale=regime | Regime stack=True
- Runtime safety: fast_mode=False | budget_triggered=histgb_bag_auto_reduce, max_fits
```

1B. Runtime Bottleneck And Fixes

Runtime bottleneck and fixes
- Profile-first instrumentation added for hotspot phase totals, per-family _predict_family timing, model fit counts, and top slow events.
- Shared split-level matrix/weight cache keyed by train/val index signatures + feature signature + half-life to eliminate repeated _xy/time-weight prep.
- Bundled family prediction path prepares X/y/w once per split and fits Ridge/Huber/HistGB/GBR/HistGB-bag/q50 sequentially.
- HistGB bagging reduced to 4 models by default (fast mode uses 1), with deterministic seed schedule and budget-aware auto-reduction.
- Candidate scan uses conservative early pruning: coarse score on first inner split, then full evaluation on top candidates only.
- Budget enforcement: total time, scan time, and fit cap stop further scanning/training and continue with best-so-far.
- Budgets triggered: histgb_bag_auto_reduce, max_fits
- Fast mode=False; optimization toggles enabled=True.
- Reproducibility: seed=23, OMP_NUM_THREADS=, MKL_NUM_THREADS=.
- Artifact validation: predictions rows=75 (expected 75), rankings rows=165 (expected 165), ranks unique 1..165=True.

1C. Fast Profiling Before vs After (timing / fits) (1/1)

kind	name	before_sec	after_sec	delta_sec
phase	_scan_and_select_core_candidates	5.230	5.524	0.294
phase	_core_scan_score	5.226	5.516	0.290
phase	_generate_inner_oof_and_outer_preds	5.540	3.973	-1.567
phase	_predict_family	10.500	9.416	-1.084
phase	_build_split_tables	0.447	0.334	-0.113
phase	_prepare_variant_outer_data	0.550	0.440	-0.110
fit_count	histgb	48.000	51.000	3.000
fit_count	histgb_bag	8.000	6.000	-2.000
fit_count	histgb_q50	8.000	6.000	-2.000
fit_count	huber	8.000	6.000	-2.000
fit_count	ridge	48.000	51.000	3.000

1D. Runtime Focus Phase Timing Summary (1/1)

phase	total_sec	calls
_scan_and_select_core_candidates	121.786	6
_core_scan_score	121.571	10320
_predict_family	102.666	54152
_prepare_variant_outer_data	5.121	24
_build_split_tables	4.645	96
_generate_inner_oof_and_outer_preds	0.506	96

1E. _predict_family Aggregated Timing / Fit Counts (1/1)

family	predict_time_sec	predict_calls	fit_count
gbr	37.479	13345	375
huber	36.728	13345	375
histgb	27.620	13345	375
ridge	0.824	13345	375
histgb_q50	0.009	385	0
histgb_bag	0.008	385	0
histgb_q20	0.000	1	0
histgb_q80	0.000	1	0

2-4. Changes / Leakage / Nested CV

```
2. What changed vs previous pipeline
- Nested tuning for post-model layers and key hyperparameters (Ridge alpha, HistGB grid, recency half-life).
- Regime-specific simplex stacking + regime-specific calibration + regime/pool scale correction with fallback.
- Recency features: EMA multi-alpha, last3/5/8, trend, consistency ratio.
- Upgraded Elo variants: dynamic-K, team-specific regularized home effect, inactivity decay.
- HistGB seed bagging and quantile HistGB (median, plus q20/q80 for diagnostics).
- Shift diagnostics (KS + PSI + out-of-range fraction) and OOD-triggered mitigation.

3. Data constraints and leakage controls
- All features are pregame-only and built sequentially.
- Static rating models refit inside each train split; no outer-fold leakage in postprocessing.
- Derby predictions are neutral-site (home effects zeroed for derby features).

4. Nested CV design (outer/inner schematic)
 outer_fold=1 train_n=125 val_n=193 train_end=2025-01-12 val=2025-01-23..2025-02-19
 outer_fold=2 train_n=318 val_n=205 train_end=2025-02-19 val=2025-03-03..2025-04-02
 outer_fold=3 train_n=523 val_n=151 train_end=2025-04-02 val=2025-04-14..2025-04-30
 outer_fold=4 train_n=674 val_n=102 train_end=2025-04-30 val=2025-05-13..2025-05-29
 outer_fold=5 train_n=776 val_n=164 train_end=2025-05-29 val=2025-05-30..2025-06-30
```

5. Outer-fold selections and metrics (1/1)

outer_candidate	dict_feature	leaf_l	lifted_edge	lasso	robust	stopule	mean	corr_p	mean	post	tail	distance	pos	mean	mean	mean	q50	outer	outer	outer	tail	disp	tail	outer	outer	outer	outer	disp	gini	gini	calibration	scale	q50	hetero	s_cap_
extra1	elitenet	seq	all	stack=pool	2000	3×pool	affin	0.556	6.68	7.12	3.46	0.028	1.53	0.22	0.000	False	6.70	6.70	5.16	5.57	2.46	7.23	0.000	0.522	0.000	0.000	5.09	0.000	5.16	0.000	False	pooled	pooled	0.000	0.000	True
extra2	elitenet	seq	all	stack=pool	2000	6×pool	affin	0.556	5.26	5.70	0.50	0.033	6.93	0.028	0.000	False	5.32	5.32	4.36	4.89	3.32	7.61	0.000	1.86	0.000	0.000	4.24	0.000	4.36	0.000	False	pooled	pooled	0.000	0.000	True
extra3	elitenet	seq	all	stack=pool	2000	9×pool	affin	0.556	4.82	5.39	9.81	0.080	7.61	0.025	0.000	False	4.41	4.41	8.59	1.29	0.62	6.40	0.000	1.79	0.000	0.000	8.59	0.000	8.59	0.000	False	pooled	pooled	0.000	0.000	True
extra4	elitenet	affine	extra	heteroskedastic	2000	Dregime	affin	0.700	2.73	3.86	4.99	0.132	0.24	0.166	0.000	False	3.14	3.14	0.84	3.33	6.28	3.94	0.257	3.36	0.282	2.20	0.49	0.548	0.49	0.340	True	regime	regime	0.200	0.000	False
extra5	elitenet	affine	extra	heteroskedastic	2000	Dregime	affin	0.556	1.06	2.45	0.47	0.144	3.89	0.215	0.000	False	1.96	1.96	4.66	5.70	3.53	7.89	0.112	4.15	0.368	9.50	7.13	0.202	7.17	2.510	True	regime	regime	0.000	0.000	False

5. Outer-fold RMSE/MAE table (1/1)

outer_fold	rmse	mae	bias	pred_std	actual_std	resid_std	tail_rmse	xtail_rmse	tail_bias	tail_dispersion_rat	corr
1	45.166	35.571	2.467	0.000	45.098	45.098	79.233	93.186	0.522	0.000	0.000
2	44.366	34.892	-3.325	0.000	44.241	44.241	77.618	86.835	-21.862	0.000	0.000
3	38.596	31.295	-0.623	0.000	38.591	38.591	66.401	71.908	1.791	0.000	0.000
4	40.845	33.339	6.289	22.207	40.496	40.358	63.947	69.811	-3.366	0.257	0.280
5	44.664	35.703	3.533	9.505	47.133	44.524	77.899	84.254	-4.158	0.112	0.368

6. Feature additions and recency ablations (core scan) (1/1)

feature_profile	half_life_days	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
no_extra_recency	60.000	44.789	44.890	44.625	612
no_extra_recency	21.000	44.792	44.890	44.624	612
no_extra_recency	90.000	44.794	44.891	44.626	612
no_extra_recency	35.000	44.797	44.889	44.625	612
compact_recency	21.000	44.800	44.929	44.608	615
compact_recency	35.000	44.805	44.927	44.608	615
compact_recency		44.810	44.930	44.609	615
compact_recency	60.000	44.840	44.972	44.641	595
no_extra_recency		44.927	45.018	44.757	622
full_recency	35.000	44.994	45.176	44.776	515
full_recency	60.000	44.995	45.178	44.776	515
full_recency	90.000	44.997	45.179	44.776	515
full_recency	21.000	44.998	45.178	44.776	515
full_recency		44.998	45.179	44.776	515
compact_recency	90.000	45.010	45.179	44.796	515

6. Core scan results (top rows) (1/4)

candidate_ke	lo_variat	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_s	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber_s	scan_ridge_s	scan_stagepr	ned_std_sca	scan_splits_u	scan_stage	outer_fold
l_recency|ele	lo_base_sta	full_recenc	35.000	32.000	2	40.783	40.577	32.903	41.701	34.041	42.678	34.310	41.395	33.880	-28.516	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	35.000	16.000	2	40.796	40.577	32.903	41.701	34.041	42.678	34.310	41.461	33.978	-28.516	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	90.000	32.000	2	40.805	40.632	32.987	41.723	33.772	42.678	34.306	41.390	33.872	-28.506	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	90.000	16.000	2	40.819	40.632	32.987	41.723	33.772	42.678	34.306	41.459	33.976	-28.506	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	60.000	32.000	2	40.822	40.602	32.942	41.863	33.907	42.681	34.323	41.392	33.874	-28.477	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	60.000	16.000	2	40.835	40.602	32.942	41.863	33.907	42.681	34.323	41.460	33.977	-28.477	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	35.000	8.000	2	40.839	40.577	32.903	41.701	34.041	42.678	34.310	41.675	34.143	-28.516	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	60.000	32.000	3	40.844	40.538	32.625	41.878	33.530	42.681	34.323	41.392	33.874	-29.304	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	60.000	16.000	3	40.857	40.538	32.625	41.878	33.530	42.681	34.323	41.460	33.977	-29.304	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	90.000	8.000	2	40.862	40.632	32.987	41.723	33.772	42.678	34.306	41.677	34.141	-28.506	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	60.000	8.000	2	40.879	40.602	32.942	41.863	33.907	42.681	34.323	41.676	34.141	-28.477	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	35.000	4.000	2	40.897	40.577	32.903	41.701	34.041	42.678	34.310	41.965	34.083	-28.516	1	coarse	1
act_recency	lo_base_sta	compact_rece	21.000	32.000	2	40.900	40.629	32.979	41.422	33.294	43.466	34.761	41.382	33.874	-28.526	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	60.000	8.000	3	40.901	40.538	32.625	41.878	33.530	42.681	34.323	41.676	34.141	-29.304	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	21.000	32.000	3	40.914	40.489	32.566	42.298	34.164	42.687	34.323	41.400	33.888	-29.325	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	90.000	4.000	2	40.921	40.632	32.987	41.723	33.772	42.678	34.306	41.970	34.080	-28.506	1	coarse	1
l_recency|ele	lo_base_sta	full_recenc	21.000	16.000	3	40.927	40.489	32.566	42.298	34.164	42.687	34.323	41.464	33.979	-29.325	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	60.000	4.000	2	40.937	40.602	32.942	41.863	33.907	42.681	34.323	41.968	34.081	-28.477	1	coarse	1
act_recency	lo_base_sta	compact_rece	21.000	16.000	2	40.943	40.629	32.979	41.422	33.294	43.466	34.761	41.597	34.110	-28.526	1	coarse	1
recency|ele	lo_base_sta	full_recenc		32.000	2	40.950	40.627	32.984	42.459	34.810	42.685	34.320	41.387	33.867	-28.480	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	35.000	2.000	2	40.956	40.577	32.903	41.701	34.041	42.678	34.310	42.262	33.992	-28.516	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	60.000	4.000	3	40.959	40.538	32.625	41.878	33.530	42.681	34.323	41.968	34.081	-29.304	1	coarse	1
recency|ele	lo_base_sta	full_recenc		16.000	2	40.964	40.627	32.984	42.459	34.810	42.685	34.320	41.458	33.976	-28.480	1	coarse	1
ll_recency|ele	lo_base_sta	full_recenc	21.000	8.000	3	40.969	40.489	32.566	42.298	34.164	42.687	34.323	41.673	34.147	-29.325	1	coarse	1

6. Core scan results (top rows) (2/4)

candidate_ke	lo_variat	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_s	scan_gbr_sc	scan_gbr_s	scan_huber_	scan_huber_s	scan_ridge_s	scan_stagepr	ned_std_sca	scan_splits_u	scan_stage	outer_fold
l_recency|he	lo_base_st	full_recenc	35.000	32.000	3	40.975	40.463	32.556	42.656	34.418	42.678	34.310	41.395	33.880	-29.346	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	90.000	2.000	2	40.980	40.632	32.987	41.723	33.772	42.678	34.306	42.267	33.994	-28.506	1	coarse	1
act_recency	lo_base_st	compact_rece	60.000	32.000	2	40.984	40.602	32.942	41.907	34.053	43.466	34.763	41.375	33.864	-28.477	1	coarse	1
l_recency|he	lo_base_st	full_recenc	35.000	16.000	3	40.988	40.463	32.556	42.656	34.418	42.678	34.310	41.461	33.978	-29.346	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	60.000	2.000	2	40.996	40.602	32.942	41.863	33.907	42.681	34.323	42.266	33.993	-28.477	1	coarse	1
recency|he	lo_base_st	full_recenc		32.000	3	41.004	40.520	32.612	42.726	34.443	42.685	34.320	41.387	33.867	-29.249	1	coarse	1
_recency|he	lo_base_st	full_recenc		8.000	2	41.008	40.627	32.984	42.459	34.810	42.685	34.320	41.678	34.139	-28.480	1	coarse	1
recency|he	lo_base_st	full_recenc		16.000	3	41.018	40.520	32.612	42.726	34.443	42.685	34.320	41.458	33.976	-29.249	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	60.000	2.000	3	41.018	40.538	32.625	41.878	33.530	42.681	34.323	42.266	33.993	-29.304	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	21.000	4.000	3	41.027	40.489	32.566	42.298	34.164	42.687	34.323	41.960	34.087	-29.325	1	coarse	1
l_recency|he	lo_base_st	full_recenc	21.000	32.000	2	41.028	40.629	32.979	42.823	34.879	42.687	34.323	41.400	33.888	-28.526	1	coarse	1
act_recency	lo_base_st	compact_rece	60.000	16.000	2	41.029	40.602	32.942	41.907	34.053	43.466	34.763	41.599	34.111	-28.477	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	35.000	8.000	3	41.031	40.463	32.556	42.656	34.418	42.678	34.310	41.675	34.143	-29.346	1	coarse	1
pact_recency	lo_base_st	compact_rece	21.000	8.000	2	41.035	40.629	32.979	41.422	33.294	43.466	34.761	42.056	34.414	-28.526	1	coarse	1
l_recency|he	lo_base_st	full_recenc	21.000	16.000	2	41.041	40.629	32.979	42.823	34.879	42.687	34.323	41.464	33.979	-28.526	1	coarse	1
_recency|he	lo_base_st	full_recenc		8.000	3	41.062	40.520	32.612	42.726	34.443	42.685	34.320	41.678	34.139	-29.249	1	coarse	1
_recency|he	lo_base_st	full_recenc		4.000	2	41.067	40.627	32.984	42.459	34.810	42.685	34.320	41.973	34.078	-28.480	1	coarse	1
act_recency	lo_base_st	compact_rece	60.000	32.000	3	41.080	40.538	32.625	42.290	33.796	43.466	34.763	41.375	33.864	-29.304	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	21.000	8.000	2	41.083	40.629	32.979	42.823	34.879	42.687	34.323	41.673	34.147	-28.526	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	21.000	2.000	3	41.086	40.489	32.566	42.298	34.164	42.687	34.323	42.256	33.990	-29.325	1	coarse	1
ll_recency|he	lo_base_st	full_recenc	35.000	4.000	3	41.089	40.463	32.556	42.656	34.418	42.678	34.310	41.965	34.083	-29.346	1	coarse	1
act_recency	lo_base_st	compact_rece	21.000	32.000	3	41.113	40.489	32.566	42.528	34.326	43.466	34.761	41.382	33.874	-29.325	1	coarse	1
_recency|he	lo_base_st	full_recenc		4.000	3	41.121	40.520	32.612	42.726	34.443	42.685	34.320	41.973	34.078	-29.249	1	coarse	1
pact_recency	lo_base_st	compact_rece	60.000	8.000	2	41.123	40.602	32.942	41.907	34.053	43.466	34.763	42.069	34.412	-28.477	1	coarse	1

6. Core scan results (top rows) (3/4)

candidate_ke	lo_variat	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_	scan_gbr_sc	scan_gbr_sc	scan_huber_	scan_huber_	scan_ridge_s	scan_stagepr	ned_std_sca	scan_splits_u	scan_stage	outer_fold
act_recency	lo_base_sta	compact_rece	60.000	16.000	3	41.124	40.538	32.625	42.290	33.796	43.466	34.763	41.599	34.111	-29.304	1	coarse	1
_recency|hle	lo_base_sta	full_recenc		2.000	2	41.127	40.627	32.984	42.459	34.810	42.685	34.320	42.271	33.995	-28.480	1	coarse	1
ll_recency|ho	lo_base_sta	full_recenc	21.000	4.000	2	41.140	40.629	32.979	42.823	34.879	42.687	34.323	41.960	34.087	-28.526	1	coarse	1
pact_recency	lo_base_sta	compact_rece	21.000	4.000	2	41.145	40.629	32.979	41.422	33.294	43.466	34.761	42.608	34.442	-28.526	1	coarse	1
ll_recency|ho	lo_base_sta	full_recenc	35.000	2.000	3	41.148	40.463	32.556	42.656	34.418	42.678	34.310	42.262	33.992	-29.346	1	coarse	1
act_recency	lo_base_sta	compact_rece	21.000	16.000	3	41.156	40.489	32.566	42.528	34.326	43.466	34.761	41.597	34.110	-29.325	1	coarse	1
_recency|hle	lo_base_sta	full_recenc		2.000	3	41.181	40.520	32.612	42.726	34.443	42.685	34.320	42.271	33.995	-29.249	1	coarse	1
act_recency	lo_base_sta	compact_rece	90.000	32.000	2	41.194	40.632	32.987	42.899	34.805	43.465	34.750	41.374	33.862	-28.506	1	coarse	1
ll_recency|ho	lo_base_sta	full_recenc	21.000	2.000	2	41.199	40.629	32.979	42.823	34.879	42.687	34.323	42.256	33.990	-28.526	1	coarse	1
pact_recency	lo_base_sta	compact_rece	60.000	8.000	3	41.218	40.538	32.625	42.290	33.796	43.466	34.763	42.069	34.412	-29.304	1	coarse	1
pact_recency	lo_base_sta	compact_rece	60.000	4.000	2	41.235	40.602	32.942	41.907	34.053	43.466	34.763	42.627	34.440	-28.477	1	coarse	1
act_recency	lo_base_sta	compact_rece	90.000	16.000	2	41.239	40.632	32.987	42.899	34.805	43.465	34.750	41.599	34.111	-28.506	1	coarse	1
l_recency|hle	lo_base_sta	full_recenc	90.000	32.000	3	41.240	40.508	32.613	43.909	35.772	42.678	34.306	41.390	33.872	-29.329	1	coarse	1
pact_recency	lo_base_sta	compact_rece	21.000	2.000	2	41.247	40.629	32.979	41.422	33.294	43.466	34.761	43.114	34.368	-28.526	1	coarse	1
pact_recency	lo_base_sta	compact_rece	21.000	8.000	3	41.248	40.489	32.566	42.528	34.326	43.466	34.761	42.056	34.414	-29.325	1	coarse	1
act_recency	lo_base_sta	compact_rece	35.000	32.000	3	41.249	40.463	32.556	43.250	34.960	43.468	34.769	41.377	33.868	-29.346	1	coarse	1
act_recency	lo_base_sta	compact_rece	35.000	32.000	2	41.253	40.577	32.903	43.278	35.503	43.468	34.769	41.377	33.868	-28.516	1	coarse	1
l_recency|hle	lo_base_sta	full_recenc	90.000	16.000	3	41.254	40.508	32.613	43.909	35.772	42.678	34.306	41.459	33.976	-29.329	1	coarse	1
act_recency	lo_base_sta	compact_rece	35.000	16.000	3	41.293	40.463	32.556	43.250	34.960	43.468	34.769	41.598	34.111	-29.346	1	coarse	1
act_recency	lo_base_sta	compact_rece	35.000	16.000	2	41.297	40.577	32.903	43.278	35.503	43.468	34.769	41.598	34.111	-28.516	1	coarse	1
ll_recency|ho	lo_base_sta	full_recenc	90.000	8.000	3	41.297	40.508	32.613	43.909	35.772	42.678	34.306	41.677	34.141	-29.329	1	coarse	1
ct_recency	lo_base_sta	compact_rece		32.000	2	41.300	40.627	32.984	43.424	35.443	43.485	34.781	41.372	33.858	-28.480	1	coarse	1
act_recency	lo_base_sta	compact_rece	90.000	32.000	3	41.324	40.508	32.613	43.558	35.527	43.465	34.750	41.374	33.862	-29.329	1	coarse	1
pact_recency	lo_base_sta	compact_rece	60.000	4.000	3	41.330	40.538	32.625	42.290	33.796	43.466	34.763	42.627	34.440	-29.304	1	coarse	1

6. Core scan results (top rows) (4/4)

candidate_k	elo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_scor	scan_histgb	scan_histgb	scan_gbr_rs	scan_gbr_sc	scan_huber	scan_huber	scan_ridge	scan_stagepred_std_sca	scan_splits_u	scan_stage	outer_fold
pact_recency	elo_base_sta	impact_rece	90.000	8.000	2	41.333	40.632	32.987	42.899	34.805	43.465	34.750	42.071	34.412	-28.506	1	coarse	1
pact_recency	elo_base_sta	impact_rece	60.000	2.000	2	41.336	40.602	32.942	41.907	34.053	43.466	34.763	43.132	34.378	-28.477	1	coarse	1
ct_recency	elo_base_sta	impact_rece		32.000	3	41.343	40.520	32.612	43.636	35.298	43.485	34.781	41.372	33.858	-29.249	1	coarse	1
ct_recency	elo_base_sta	impact_rece		16.000	2	41.345	40.627	32.984	43.424	35.443	43.485	34.781	41.600	34.112	-28.480	1	coarse	1
ll_recency	elo_base_sta	full_recenc	90.000	4.000	3	41.356	40.508	32.613	43.909	35.772	42.678	34.306	41.970	34.080	-29.329	1	coarse	1
pact_recency	elo_base_sta	impact_rece	21.000	4.000	3	41.358	40.489	32.566	42.528	34.326	43.466	34.761	42.608	34.442	-29.325	1	coarse	1
act_recency	elo_base_sta	impact_rece	90.000	16.000	3	41.369	40.508	32.613	43.558	35.527	43.465	34.750	41.599	34.111	-29.329	1	coarse	1
pact_recency	elo_base_sta	impact_rece	35.000	8.000	3	41.386	40.463	32.556	43.250	34.960	43.468	34.769	42.064	34.413	-29.346	1	coarse	1

7. Elo variant comparisons (1/1)

elo_variant	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
elo_dynamic_k_teamha	44.650	44.667	44.667	1875
elo_dynamic_k_teamha_decay	44.650	44.667	44.667	1875
elo_dynamic_k	44.696	44.711	44.711	1885
elo_base_static	45.296	45.653	44.698	2965

7. HistGB hyperparameter sweep summary (1/1)

histgb_idx	scan_score_mean	scan_histgb_rmse_mean	n	params
3	44.818	44.755	1753	{'learning_rate': 0.025, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 5
2	44.812	44.760	1753	{'learning_rate': 0.031, 'max_depth': 4, 'max_leaf_nodes': 63, 'min_samples_leaf': 6,
4	44.847	44.853	1738	{'learning_rate': 0.061, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 12
1	44.708	44.861	1678	{'learning_rate': 0.051, 'max_depth': 4, 'max_leaf_nodes': 31, 'min_samples_leaf': 8,
0	45.234	45.884	1678	{'learning_rate': 0.041, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10

8. Regime-specific stacking / calibration results (1/1)

module_name	use_regime_sta	calibration_mod	scale_mode	eta	outer_rmse_mo	outer_mae_mo	outer_tail_dispersion	outer_tail_bias_m	outer_corr_mo	outer_dispersion_	regime_mi	gain_vs_poole	n
regime_affine_v	True	regime	regime	0.700	40.845	33.339	63.947	0.257	-3.366	0.280	0.548	-0.346	16
xpand_affine_gl	False	pooled	pooled	0.550	42.709	33.919	74.417	0.000	-6.516	0.000	0.000	0.000	48
regime_affine_v	True	regime	regime	0.550	44.664	35.703	77.899	0.112	-4.158	0.368	0.202	2.510	16

9. Dispersion and scale correction analysis (1/1)

outer_fold	stage	pred_std	actual_std	rmse
1	raw_mean	0.000	45.098	45.241
1	calibrated	0.000	45.098	87.250
1	scaled	0.000	45.098	67.838
1	final	0.000	45.098	45.166
2	raw_mean	0.000	44.241	44.274
2	calibrated	0.000	44.241	44.268
2	scaled	0.000	44.241	44.268
2	final	0.000	44.241	44.366
3	raw_mean	0.000	38.591	38.597
3	calibrated	0.000	38.591	39.006
3	scaled	0.000	38.591	38.662
3	final	0.000	38.591	38.596
4	raw_mean	0.000	40.496	40.528
4	calibrated	18.371	40.496	43.113
4	scaled	16.647	40.496	40.573
4	final	22.207	40.496	40.845
5	raw_mean	0.000	47.133	47.137
5	calibrated	15.975	47.133	44.128
5	scaled	11.182	47.133	44.286
5	final	9.505	47.133	44.664

10. Shift diagnostics: Train recent vs Derby (1/1)

feature	train_recent_mean	derby_mean	train_recent_std	derby_std	ks_stat	ks_pvalue	psi	derby_outside_train_1_9	ood_flag
games_played_min	9.122	11.067	1.095	0.660	0.730	0.000	8.252	0.227	True
elo_diff_pre	58.187	-0.325	64.506	58.625	0.409	0.000	2.070	0.107	True
same_conf_flag	0.697	0.013	0.459	0.115	0.684	0.000	0.000	0.000	True
massey_diff	5.872	2.195	19.548	15.187	0.166	0.075	0.192	0.000	False
consistency_ratio_di	0.059	-0.008	0.657	0.577	0.113	0.427	0.167	0.013	False
_margin_last5_vs_sea	-1.434	1.554	16.927	19.606	0.105	0.514	0.133	0.067	False
volatility_sum	71.164	73.217	11.169	9.954	0.139	0.196	0.132	0.027	False
offdef_net_diff	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
offdef_margin_neutr	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
elo_neutral_diff_pre	3.187	-0.325	64.506	58.625	0.099	0.591	0.053	0.013	False

10. Shift Mitigation Notes

10. Shift mitigation rules
- Detected 3 OOD-like feature shifts (KS/PSI/out-of-range checks).
- Applied softer scaling (blend toward 1.0) and wider winsor bounds under OOD risk.
- No anti-compression guard applied (derby std 36.35, ref 17.04).

8. Regime weights (outer_fold_1_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_huber	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	103	0.200	0.200	0.200	0.200	0.200

8. Regime weights (outer_fold_2_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_huber	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	284	0.200	0.200	0.200	0.200	0.200

8. Regime weights (outer_fold_3_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_huber	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	448	0.200	0.200	0.200	0.200	0.200

8. Regime weights (outer_fold_4_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_huber	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	549	0.200	0.200	0.200	0.200	0.200
full	moderate|same|lowinfo|low	87	0.200	0.200	0.200	0.200	0.200
full	moderate|same|lowinfo|hig	58	0.200	0.200	0.200	0.200	0.200
simple	moderate|same|lowinfo	145	0.200	0.200	0.200	0.200	0.200
simple	moderate|cross|lowinfo	62	0.200	0.200	0.200	0.200	0.200
simple	close|same|lowinfo	60	0.200	0.200	0.200	0.200	0.200
simple	moderate|same|midinfo	67	0.200	0.200	0.200	0.200	0.200
simple	close|same|midinfo	54	0.200	0.200	0.200	0.200	0.200
simple	mismatch|same|midinfo	45	0.200	0.200	0.200	0.200	0.200
gap	moderate	305	0.200	0.200	0.200	0.200	0.200
gap	close	144	0.200	0.200	0.200	0.200	0.200
gap	mismatch	100	0.200	0.200	0.200	0.200	0.200

8. Regime weights (outer_fold_5_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_huber	w_pred_histgb	w_pred_gbr	w_pred_histgb_bag
pooled	ALL	589	0.200	0.200	0.200	0.200	0.200
full	moderate|same|lowinfo|low	77	0.200	0.200	0.200	0.200	0.200
full	moderate|same|lowinfo|hig	51	0.200	0.200	0.200	0.200	0.200
full	moderate|same|midinfo|hig	45	0.200	0.200	0.200	0.200	0.200
simple	moderate|same|lowinfo	128	0.200	0.200	0.200	0.200	0.200
simple	close|same|lowinfo	57	0.200	0.200	0.200	0.200	0.200
simple	moderate|same|midinfo	72	0.200	0.200	0.200	0.200	0.200
simple	close|same|midinfo	57	0.200	0.200	0.200	0.200	0.200
simple	mismatch|same|midinfo	50	0.200	0.200	0.200	0.200	0.200
gap	moderate	288	0.200	0.200	0.200	0.200	0.200
gap	close	170	0.200	0.200	0.200	0.200	0.200
gap	mismatch	131	0.200	0.200	0.200	0.200	0.200

8. Calibration params (outer_fold_1_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	103	77.365	0.300

8. Calibration params (outer_fold_2_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	284	-3.686	1.063

8. Calibration params (outer_fold_3_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	448	9.285	0.300

8. Calibration params (outer_fold_4_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	549	5.098	0.300
full	moderate|same|lowinfo|lowvol	87	8.899	0.300
full	moderate|cross|lowinfo|highvol	36	19.604	0.945
full	moderate|same|lowinfo|highvol	58	6.266	0.300
full	close|same|lowinfo|lowvol	44	23.736	0.300
full	moderate|same|midinfo|highvol	44	-12.430	2.200
simple	moderate|same|lowinfo	145	11.017	0.300
simple	moderate|cross|lowinfo	62	126.864	0.300
simple	close|same|lowinfo	60	8.308	0.300
simple	mismatch|same|lowinfo	33	31.426	0.735
simple	moderate|same|midinfo	67	-33.026	2.200
simple	close|same|midinfo	54	-77.857	2.200
simple	mismatch|same|midinfo	45	5.011	2.200
gap	moderate	305	-11.027	2.200
gap	close	144	17.090	0.300
gap	mismatch	100	24.514	0.883

8. Calibration params (outer_fold_5_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	589	11.008	0.300
full	moderate|same|lowinfo|lowvol	77	-52.929	2.200
full	close|same|lowinfo|lowvol	43	-187.345	2.200
full	moderate|same|lowinfo|highvol	51	-47.618	2.200
full	moderate|same|midinfo|highvol	45	-3.763	1.718
simple	moderate|same|lowinfo	128	-47.337	2.200
simple	close|same|lowinfo	57	-74.422	2.200
simple	moderate|cross|lowinfo	29	169.338	0.300
simple	mismatch|same|lowinfo	32	31.109	0.870
simple	moderate|same|midinfo	72	-5.273	1.650
simple	close|same|midinfo	57	-27.406	2.200
simple	mismatch|same|midinfo	50	88.780	0.300
simple	moderate|cross|midinfo	31	53.121	0.300
gap	moderate	288	-2.547	1.090
gap	close	170	-6.747	0.300
gap	mismatch	131	25.914	0.832

9. Scale params (outer_fold_1_selected) (1/1)

level	regime	n	scale
pooled	ALL	103	0.700

9. Scale params (outer_fold_2_selected) (1/1)

level	regime	n	scale
pooled	ALL	284	1.000

9. Scale params (outer_fold_3_selected) (1/1)

level	regime	n	scale
pooled	ALL	448	0.700

9. Scale params (outer_fold_4_selected) (1/1)

level	regime	n	scale
pooled	ALL	549	0.700
full	moderate|same|lowinfo|lowvol	87	0.700
full	moderate|cross|lowinfo|highvol	36	1.000
full	moderate|same|lowinfo|highvol	58	0.700
full	close|same|lowinfo|lowvol	44	0.700
full	moderate|same|midinfo|highvol	44	1.650
simple	moderate|same|lowinfo	145	0.700
simple	moderate|cross|lowinfo	62	0.700
simple	close|same|lowinfo	60	0.700
simple	mismatch|same|lowinfo	33	1.000
simple	moderate|same|midinfo	67	0.700
simple	close|same|midinfo	54	0.700
simple	mismatch|same|midinfo	45	1.300
gap	moderate	305	0.700
gap	close	144	0.700
gap	mismatch	100	1.100

9. Scale params (outer_fold_5_selected) (1/1)

level	regime	n	scale
pooled	ALL	589	0.700
full	moderate|same|lowinfo|lowvol	77	0.700
full	close|same|lowinfo|lowvol	43	0.700
full	moderate|same|lowinfo|highvol	51	0.700
full	moderate|same|midinfo|highvol	45	1.000
simple	moderate|same|lowinfo	128	0.700
simple	close|same|lowinfo	57	0.700
simple	moderate|cross|lowinfo	29	0.700
simple	mismatch|same|lowinfo	32	1.000
simple	moderate|same|midinfo	72	0.850
simple	close|same|midinfo	57	0.900
simple	mismatch|same|midinfo	50	0.700
simple	moderate|cross|midinfo	31	0.700
gap	moderate	288	0.700
gap	close	170	0.700
gap	mismatch	131	0.700

11. Residual Diagnostics (Summary)

```
11. Residual diagnostics
- OOF final RMSE=43.178, MAE=34.355, Bias=1.130
- Residual std=43.164, skew=-0.070, kurtosis=-0.205
- Mean-stack RMSE=43.663
- Quantile median (q50) RMSE=43.663
- Mean+median blend RMSE=43.663
- Calibrated RMSE=56.586
- Tail error threshold (top 10%)=70.568
```

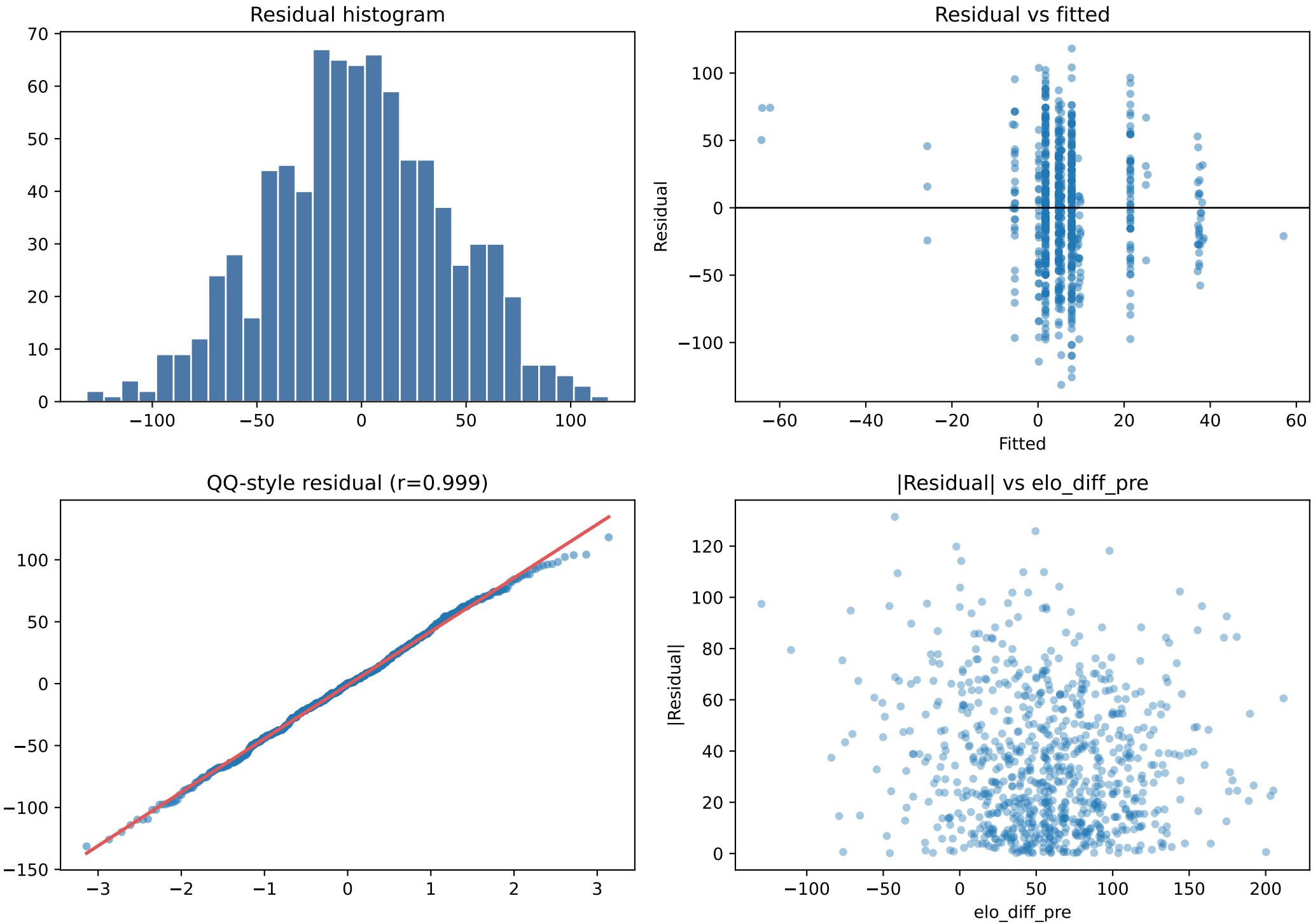
11. Tail error diagnostics (top 10% abs-error) (1/2)

Date	elo_diff	passes	off_def	volatility	volatility_squared	pass_plays	pass_rate	rush_rate	comeback	blowout	consistency	row_index	true	pred_raw	pred_hist	pred_cal	pred_sigmoid	pred_isotonic	pred_ensemble	pred_hist_hi	pred_cal_hi	pred_sigmoid_hi	pred_isotonic_hi	pred_ensemble_hi	pred_cal_lo	pred_sigmoid_lo	pred_isotonic_lo	pred_ensemble_lo	pred_cal_mid	pred_sigmoid_mid	pred_isotonic_mid	pred_ensemble_mid	calibrator	model	abs_error
-06-16 0	-42.44	28.50	-42.00	34.46	34.37	10.000	10.000	1.000	0.000	15.400	16.480	-0.336	899	-126.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	131.39
-02-05 0	49.528	5.428	-5.714	15.00	12.754	2.000	3.000	1.000	0.000	0.000	-1.387	236	-118.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	125.82
-02-19 0	-2.217	-0.798	-8.000	10.625	9.708	3.000	4.000	1.000	0.000	0.000	0.000	0.717	267	-112.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	119.82
-02-19 0	97.907	12.251	6.286	33.893	31.000	4.000	2.000	1.000	0.000	0.000	0.000	1.510	269	126.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	118.17
-06-30 0	0.973	15.186	19.896	54.529	26.609	11.000	10.000	0.000	1.000	-3.582	-3.755	0.377	829	-114.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	114.19
-01-23 0	55.117	1.724	-6.750	30.000	9.000	2.000	2.000	1.000	0.000	0.000	0.000	-1.094	134	-102.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	109.82
-02-05 0	41.541	2.175	-5.000	29.394	13.000	3.000	2.000	1.000	0.000	0.000	0.000	-3.571	196	-102.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	109.82
-06-16 0	-40.62	4.844	26.282	4.985	41.395	11.000	11.000	0.000	1.000	11.745	13.361	1.457	907	-104.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	109.39
-01-24 0	55.116	0.461	5.429	45.000	45.000	1.000	0.000	0.000	1.000	0.000	0.000	0.826	176	112.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	104.17
-06-30 0	0.257	-6.552	6.219	2.571	29.441	8.000	9.000	1.000	0.000	-12.06	-11.93	1.381	804	104.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	103.80
-04-02 0	43.90	1.036	9.782	43.916	34.006	5.000	7.000	1.000	0.000	-0.857	-1.214	0.386	488	104.00	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	exp_an	poole	102.26
-01-23 0	44.757	-4.073	5.643	2.000	45.000	2.000	1.000	0.000	1.000	0.000	0.000	1.870	128	-94.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	101.82
-01-24 0	34.426	-0.510	2.284	5.000	45.000	1.000	1.000	0.000	1.000	0.000	0.000	1.261	139	-94.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	101.82
-03-04 0	14.546	5.403	-0.022	5.281	25.020	4.000	4.000	1.000	0.000	0.000	0.000	-0.107	385	100.00	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	exp_an	poole	98.262
-03-03 0	31.552	0.049	-5.200	7.864	28.914	4.000	4.000	1.000	0.000	0.000	0.000	-0.165	325	-96.00	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	exp_an	poole	97.738
-06-15 0	21.366	6.736	3.143	5.584	4.463	8.000	8.000	0.000	1.000	-43.20	-43.01	0.061	697	-88.00	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	exp_an	logistic_reg	97.559
-06-30 0	129.61	42.205	3.125	7.552	2.428	9.000	10.000	1.000	0.000	5.533	6.102	0.857	799	-76.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	97.448
-06-14 0	-46.00	28.029	5.306	3.343	2.450	10.000	8.000	0.000	1.000	37.000	37.984	-0.010	687	-102.00	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	5.953	exp_an	logistic_reg	96.591
-06-16 0	58.382	2.158	1.600	0.669	3.457	10.000	10.000	1.000	0.000	38.800	38.931	-1.679	903	118.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	96.552
-06-27 0	0.018	-4.515	3.125	9.976	2.180	10.000	11.000	0.000	1.000	18.690	16.859	-0.354	928	-96.00	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_an	logistic_reg	96.196
-02-19 0	56.345	0.396	2.250	3.795	6.949	3.000	3.000	1.000	0.000	0.000	0.000	0.499	304	104.00	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	poole	96.176
-04-02 0	54.229	5.403	-0.800	2.007	0.529	4.000	6.000	1.000	0.000	11.200	11.185	-0.191	491	-94.00	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	6.767	exp_an	poole	95.738

11. Tail error diagnostics (top 10% abs-error) (2/2)

Date	elo_diff	massey	offdef	volatility	volume	days_since	days_since_prev	home_adv	neutral	home_adv_eff	pace_5	pace_consistency	row_idx	y_true	used_prior	used_prior_home	used_prior_hist	pred_raw	pred_hist	pred_cal	pred_calibrated	pred_source	pred_fallback	pred_raw	pred_raw_home	pred_raw_hist	pred_cal	pred_calibrated	pred_source	pred_fallback	abs_error
-05-16 0	56.772	1.919	0.933	6.120	3.926	9.000	9.000	1.000	0.000	7.511	7.101	0.430	718	90.000	5.953	5.953	5.953	5.953	5.953	5.953	5.953	exp_ana	1	5.953	5.953	5.953	5.953	5.953	exp_ana	1	95.393
-04-30 0	71.266	6.493	6.000	0.786	3.790	8.000	8.000	1.000	0.000	11.000	11.683	1.659	621	-90.000	6.099	6.099	6.099	6.099	6.099	6.099	6.099	module	3	6.099	6.099	6.099	6.099	6.099	module	3	94.821
-03-04 0	72.641	4.183	6.626	6.106	4.500	5.000	3.000	1.000	0.000	0.000	0.000	0.923	365	96.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	94.262
-03-04 0	7.705	-6.828	2.222	8.219	1.776	3.000	3.000	1.000	0.000	0.000	0.000	3.604	342	-92.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	93.738
-06-16 0	74.543	6.803	8.029	7.250	8.077	9.000	10.000	1.000	0.000	-10.800	11.661	-0.041	890	114.000	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_ana	5	5.740	5.740	5.740	5.740	5.740	exp_ana	5	92.552
-04-02 0	38.114	4.612	2.444	1.784	2.464	5.000	5.000	0.000	1.000	0.000	0.000	1.407	495	94.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	92.262
-02-19 0	32.233	3.797	-5.214	8.380	2.881	3.000	3.000	1.000	0.000	0.000	0.000	-1.325	294	-82.000	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	1	8.944	8.944	8.944	8.944	8.944	module	1	89.824
-03-03 0	31.714	0.724	-7.271	1.214	2.032	5.000	3.000	1.000	0.000	0.000	0.000	1.248	324	-88.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	89.738
-03-17 0	23.144	-8.498	5.581	5.461	8.262	5.000	4.000	1.000	0.000	0.000	0.000	2.786	393	90.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	88.262
-03-18 0	93.021	2.425	8.306	3.952	5.826	3.000	4.000	1.000	0.000	0.000	0.000	-1.559	428	90.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	88.262
-04-02 0	18.622	3.156	6.200	6.569	5.710	6.000	6.000	1.000	0.000	-1.067	-1.288	-1.087	494	90.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	88.262
-04-15 0	55.562	1.844	2.333	6.559	2.461	6.000	7.000	0.000	1.000	5.819	5.508	-1.542	539	92.000	6.099	6.099	6.099	6.099	6.099	6.099	6.099	module	3	6.099	6.099	6.099	6.099	6.099	module	3	87.179
-04-14 0	-14.38	-3.183	0.545	4.397	8.493	5.000	5.000	1.000	0.000	0.000	0.000	0.421	525	-82.000	6.099	6.099	6.099	6.099	6.099	6.099	6.099	module	3	6.099	6.099	6.099	6.099	6.099	module	3	86.821
-04-01 0	59.493	0.254	4.500	4.011	2.214	4.000	5.000	1.000	0.000	0.000	0.000	-0.141	457	88.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	86.262
-01-23 0	12.562	7.453	6.254	2.279	5.000	3.000	1.000	1.000	0.000	0.000	0.000	1.817	129	-78.000	8.944	8.944	8.944	8.944	8.944	8.944	8.944	module	1	8.944	8.944	8.944	8.944	8.944	module	1	85.824
-03-18 0	9.586	2.724	9.092	7.457	5.010	6.000	6.000	1.000	0.000	-3.200	-3.099	-1.229	432	-84.000	6.767	6.767	6.767	6.767	6.767	6.767	6.767	module	2	6.767	6.767	6.767	6.767	6.767	module	2	85.738
-04-16 0	78.114	1.300	-0.333	1.988	6.254	6.000	6.000	0.000	1.000	0.667	0.666	0.324	542	-80.000	6.099	6.099	6.099	6.099	6.099	6.099	6.099	module	3	6.099	6.099	6.099	6.099	6.099	module	3	84.821
-06-12 0	81.092	8.105	6.152	7.690	1.404	8.000	10.000	1.000	0.000	-8.500	10.079	-0.433	851	106.000	5.740	5.740	5.740	5.740	5.740	5.740	5.740	exp_ana	5	5.740	5.740	5.740	5.740	5.740	exp_ana	5	84.552

11. Residual diagnostics plots



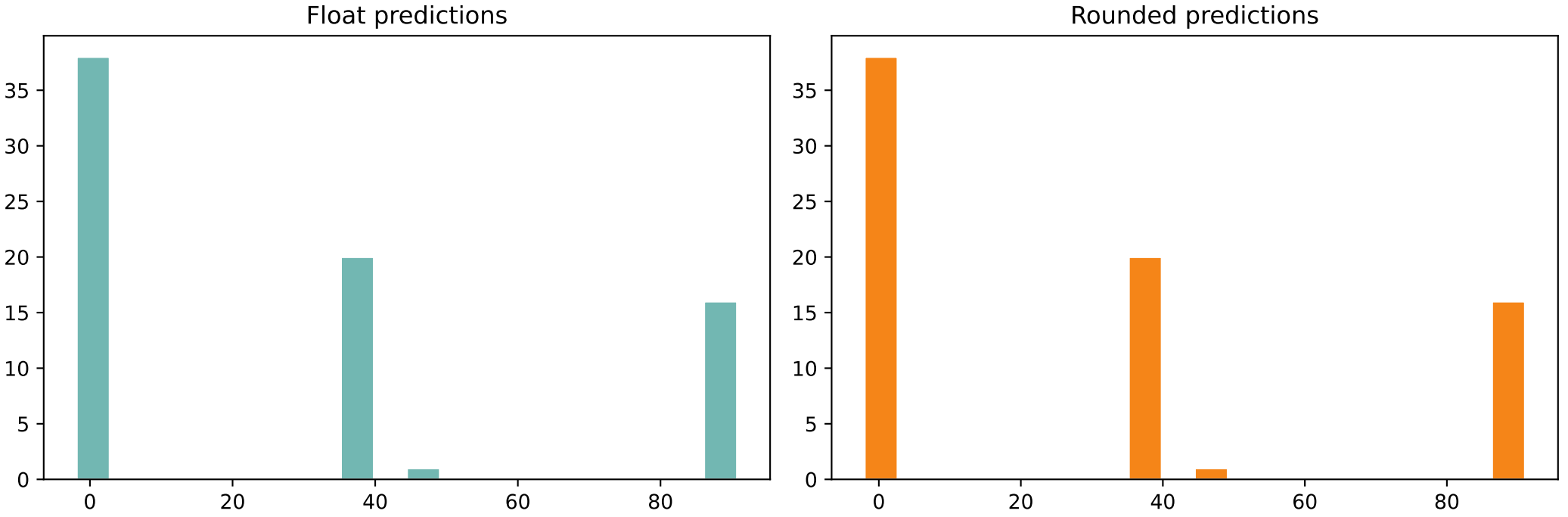
12-14. Final Decisions / Limitations / Next Steps

```
12. Final derby prediction distribution
- count=75, mean=29.191, std=36.351
- min=-1.852, max=90.740
- quantiles 1/5/25/50/75/95/99 = -1.852, -1.852, -1.852, -1.852, 38.085, 90.740, 90.740

13. Final decisions and rationale
- Final core candidate: elo_base_static|no_extra_recency|hl=none|ridge=2|hgb=0
- Final postprocess candidate: module=expand_regime_affine_v2_heterosk|stack=reg|cal=regime|scale=regime|q50=0.00|win=0.000|eta=0.55
- Recency profile=no_extra_recency | half_life=None
- Shift mitigation applied=True
- Dispersion guard applied=False

14. Limitations + next steps
- Derby labels hidden: final selection relies on nested train-only proxies.
- No injuries/lineups/travel features available.
- Future work: monotone piecewise regime calibration and richer Bayesian uncertainty.
```

12. Final derby prediction distribution



Appendix A. First 10 rows of predictions.csv (1/1)

GameID	Date	Team1_Conf	Team1_ID	Team1	Team2_Conf	Team2_ID	Team2	Team1_WinMargin
1941	7/4/2025	Purple	68	Idaho	Yellow	119	Houston	-2
1942	7/4/2025	White	105	Helena	Yellow	120	Iowa	-2
1943	7/4/2025	Red	82	Jackson	Purple	67	Concord	38
1944	7/4/2025	White	108	Montpelier	Yellow	129	West Virginia	-2
1945	7/4/2025	Yellow	121	Lansing	White	113	South Dakota	-2
1946	7/4/2025	Crimson	21	Lincoln	Yellow	118	Florida	-2
1947	7/4/2025	Yellow	117	Columbia	Crimson	23	Nashville	38
1948	7/4/2025	Orange	63	New Jersey	Yellow	126	Pierre	-2
1949	7/4/2025	Purple	77	Wilmington	Orange	58	Honolulu	-2
1950	7/4/2025	Crimson	22	Michigan	Green	49	Natchez	38

Appendix B. Top 20 rankings by Rank (1/1)

TeamID	Team	Rank
36	Massachusetts	1
90	Big Sur	2
94	Kentucky	3
37	Minnesota	4
40	Pennsylvania	5
45	Boise City	6
2	Baton Rouge	7
153	Tokyo	8
26	Phoenix	9
92	Illinois	10
88	Port St. Lucie	11
137	Milan	12
46	Boston	13
54	Boise	14
47	Chicago	15
19	Georgia	16
13	Texas	17
85	New Hope	18
131	Belgrade	19
149	Seoul	20

Appendix C. Selected hyperparameters (1/1)

name	value
elo_variant	elo_base_static
feature_profile	no_extra_recency
half_life_days	none
ridge_alpha	2.0
histgb_idx	0
histgb_params	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10, 'l2_regularization': 0.3, 'ma
postprocess	module=expand_regime_affine_v2_heterosk|stack=reg|cal=regime|scale=regime|q50=0.00|win=0.000|eta
postprocess_module	expand_regime_affine_v2_heterosk
postprocess_eta	0.55
winsor_bounds	(-116.61, 113.305)
dispersion_guard_factor	1.0
seed	23
OMP_NUM_THREADS	
MKL_NUM_THREADS	

Appendix D. Final derby stage outputs (head) (1/1)

Date	elo_diff	mass	eff	coef_var	volatility	volatility_adj	mean_age	mean_years_plays	mean_cohort	mean_cohort_adj	mean_pass5	mean_pass5_adj	consistency	row_index	pred_rind	pred_hold	pred_hist	pred_rd	pred_histgb	pred_q5	pred_q2	stage_rank	stage_rank	stage_rank	stage_cash	stage_sum	stage_mile	spread_q
-07-04 0	-4.319	-0.364	-3.797	35.596	29.274	11.000	12.000	0.000	1.000	11.839	11.198	-0.229	0	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	22.378	14.106	12.458	39.164	40.902	10.000	12.000	0.000	1.000	-20.800	21.102	0.499	1	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	61.146	-3.146	-7.864	59.498	27.316	11.000	13.000	0.000	1.000	9.849	10.915	-0.544	2	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	43.100	36.635	38.085	0.000
-07-04 0	27.445	2.010	-4.412	44.919	28.756	10.000	11.000	0.000	1.000	21.600	19.898	0.109	3	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	20.059	10.874	16.750	38.940	36.984	12.000	10.000	0.000	1.000	-5.000	-4.746	-0.546	4	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	21.386	8.446	1.765	30.409	33.738	11.000	11.000	0.000	1.000	-21.527	21.221	0.054	5	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	35.060	1.343	2.118	40.055	39.435	11.000	11.000	0.000	1.000	-2.073	-3.274	-0.082	6	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	43.100	36.635	38.085	0.000
-07-04 0	10.828	14.158	8.985	35.920	32.824	10.000	11.000	0.000	1.000	7.418	7.978	0.381	7	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	34.104	0.807	-0.588	41.956	41.415	11.000	11.000	0.000	1.000	-17.097	17.566	-0.023	8	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	66.423	13.143	21.640	28.163	37.988	10.000	11.000	0.000	1.000	12.927	14.346	1.020	9	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	43.100	36.635	38.085	0.000
-07-04 0	35.309	6.094	1.544	41.287	34.893	10.000	11.000	0.000	1.000	-3.855	-3.785	0.003	10	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	43.100	36.635	38.085	0.000
-07-04 0	11.625	-4.250	-8.824	32.105	38.441	11.000	11.000	0.000	1.000	-9.964	-9.016	-0.303	11	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	18.201	1.607	0.608	31.302	40.159	11.000	12.000	0.000	1.000	-0.827	-0.219	0.101	12	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	11.916	-2.432	-8.588	34.862	33.405	12.000	11.000	0.000	1.000	-12.909	12.470	-0.039	13	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	42.745	7.018	9.059	33.643	31.326	11.000	11.000	0.000	1.000	1.200	-0.538	0.401	14	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	43.100	36.635	38.085	0.000
-07-04 0	26.821	0.733	-3.882	33.735	36.430	11.000	11.000	0.000	1.000	19.600	20.081	-0.155	15	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	10.577	5.344	-0.026	28.877	37.659	12.000	11.000	0.000	1.000	11.533	12.246	0.065	16	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	19.333	-0.458	-2.706	58.962	38.414	11.000	12.000	0.000	1.000	41.327	40.959	-0.193	17	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	31.223	6.648	-6.941	33.206	35.506	11.000	11.000	0.000	1.000	2.327	3.044	-0.150	18	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000
-07-04 0	7.150	-3.761	-6.333	44.267	37.696	12.000	12.000	0.000	1.000	35.100	33.916	-0.273	19	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	5.853	-12.175	10.349	-1.852	0.000